NYC Yellow Taxi Trip Data Analysis

January 2025 — Final Project Report

Project Objective

The objective of this project is to perform an end-to-end exploratory data analysis of NYC Yellow Taxi trip records using Python, NumPy, Pandas, Matplotlib, and Seaborn. The analysis focuses on trip activity, distance, duration, passenger patterns, time trends, locations, payment types, vendors, rate codes, and operational data quality.

1. Dataset Description

The dataset is the NYC Yellow Taxi trip dataset for January 2025, supplied as a Parquet file. It contains trip-level operational information including pickup and drop-off timestamps, passenger count, trip distance, rate code, pickup and drop-off Taxi Zone IDs, vendor, payment type, and store-and-forward information. The dataset does not provide fare, tip, revenue, profit, customer ID, or driver ID fields, so the analysis does not make revenue or customer-spending claims.

Original dataset size: **3,475,226 trips**.

2. Data-Quality Findings

Check	Finding
Duplicate records	0 duplicate rows
Negative trip duration	124 records
Zero trip duration	1,927 records
Duration above 24 hours	20 records
Zero trip distance	90,196 records; investigated and retained
Extreme distance	116 records above 1,000 miles; clearly corrupted
Suspicious speed	High-speed records investigated before removal
Passenger count	0 and 7–9 passenger records investigated and retained as unusual
Location IDs	Pickup and drop-off IDs remained within 1–265
Categorical values	Unusual values investigated rather than automatically deleted

3. Cleaning Methodology

A separate clean_df DataFrame was created so that the original data remained unchanged. Cleaning focused on records identified as logically impossible or clearly corrupted rather than deleting statistical outliers automatically.

• Removed negative, zero-duration, and greater-than-24-hour trips because the project requires positive and realistic trip durations for duration and speed analysis.

• Investigated zero-distance trips and retained them because zero distance was not automatically treated as invalid.

• Removed trips with trip_distance > 1,000 miles after inspection showed clearly corrupted values.

• Removed 500 records with duration <= 1 minute and distance > 1 mile after inspection identified impossible combinations.

• Removed remaining records with average_speed_mph > 100 because the observed values were logically impossible for taxi trips.

• Missing categorical values were retained where there was insufficient evidence that they represented invalid records.

• IQR outliers were investigated but were not automatically removed simply because they were statistical outliers.

Cleaning result: 3,472,251 records remained, meaning approximately 99.91% of the original dataset was retained. A total of 2,975 records were removed.

4. Major Statistical Results

Metric	Result
Average trip distance	3.10 miles
Median trip distance	1.67 miles
Maximum trip distance	265.90 miles
Average trip duration	15.03 minutes
Median trip duration	11.70 minutes
Maximum trip duration	1,439.87 minutes
Average speed	11.07 mph
Median speed	9.52 mph
Most common passenger count	1 passenger
Distance-duration correlation	0.348
Distance-speed correlation	0.733

5. Major Visualizations & Insights

Trips by Pickup Hour — Demand is lowest in the early morning and rises through the day, reaching its highest level around 6 PM.

Trips by Day of Week — Thursday recorded the highest number of trips, while Monday was the lowest among the seven days.

Hour × Day Demand Heatmap — Demand is concentrated in afternoon and evening periods, with strong activity around Thursday and Friday evenings.

Trip Distance Distribution — Most trips are short, with the largest concentration around the 1–3 mile range and a long right tail.

Trip Duration Distribution — Most trips fall around the 5–15 minute range, with a right-skewed distribution caused by longer trips.

Distance vs Duration Scatter Plot — The relationship is positive but widely dispersed. The correlation of about 0.35 indicates a moderate positive relationship rather than a deterministic one.

Trips by Distance Category — Short trips form the largest group, followed by very short and medium trips.

Average Duration by Distance Category — Average duration increases as the distance category increases, showing a clear operational relationship between trip length and elapsed time.

Payment Type Distribution — Payment type 1 is the dominant category, accounting for about 70.35% of cleaned trips.

Vendor Distribution — Vendor 2 represents about 78.32% of cleaned trips, while Vendor 6 has only a very small number of records and should not be used for broad comparisons.

Top Pickup and Drop-off Locations — Location 161 is the most common pickup location and location 236 is the most common drop-off location in the cleaned dataset.

Top Routes — The most frequent route is 237 → 236 with 24,819 trips, followed by other concentrated routes involving the highest-volume zones.

6. Key Insights

• Taxi demand peaks around 6 PM and is lowest around 4 AM.

• Weekday activity is substantially higher than weekend activity in this January dataset.

• A typical cleaned trip is short: the median trip is 1.67 miles and 11.7 minutes with one passenger.

• Thursday is the busiest day by trip count.

• Longer distance categories generally have longer average trip durations.

• Passenger count has very weak relationships with trip distance and duration.

• The strongest correlation among the analyzed variables is between trip distance and average speed, while distance and duration have a moderate positive relationship.

• Payment type 1 is the most frequent payment category.

• Vendor 2 accounts for the large majority of trips, so comparisons involving very small groups such as Vendor 6 require caution.

• A small number of clearly corrupted records were removed after investigation, while statistical outliers were retained when they could represent valid unusual trips.

7. Limitations

• The analysis covers January 2025 only and may not represent other months or seasons.

• Pickup and drop-off values are Taxi Zone IDs rather than neighborhood names because no separate lookup table was provided.

• Several categorical fields contain missing values; these were retained rather than filled with unsupported assumptions.

• Some unusual but potentially valid observations remain after cleaning.

• Very small groups, such as rare vendor, payment, or rate-code categories, should not support broad conclusions.

• Correlation indicates association, not causation.

• The dataset does not support claims about revenue, profit, customer spending, driver earnings, or individual customer behavior.

8. Conclusion

This project demonstrates a complete exploratory data analysis workflow from raw Parquet data through data-quality assessment, cleaning, feature engineering, statistical analysis, visualization, and business interpretation. The cleaned January 2025 dataset shows strong concentration of taxi activity during daytime and evening hours, especially around 6 PM, with Thursday being the busiest day. Most trips are short and involve one passenger. Distance and duration show a moderate positive relationship, while passenger count has little relationship with the main trip characteristics. The cleaning process removed only clearly corrupted or logically impossible records and retained valid statistical outliers, producing a final analytical dataset of 3,472,251 trips.